

Factor 3 mostly switches the closure off

GABLS1 factor-3 sensitivity | 22 September 2026 | 12.5 m, WENO9, one interior face, 300 s filter, 9 h, seed 123. New plots compare only factors 1 and 3, with fixed 1 m LES reference medians where available. Earlier reports and data are preserved.

Result: factor 3 largely switches off the added near-surface closure after spin-up, while resolved turbulence increases strongly. In 8-9 h, first-face w^2 rises from 0.00817056 to 0.060734 m^2/s^2 , a factor of 7.433. Integrated resolved TKE increases 16.38%. First-face w^3 and skewness change from negative to positive. This is a large response to the assumed missing-transport credit, not a calibrated accuracy result.

The intervention remains narrow: multiply signed, locally time-filtered resolved flux by three only inside $\max(0, 1 - a F_{\text{resolved}}/F_{\text{target}})$, for momentum and heat. Aligned resolved transport reaching one-third of the target can eliminate that local deficit; countergradient transport increases it. The assumed extra transport is twice the resolved contribution. Coefficients respond dynamically; neither viscosity nor plotted covariance is simply multiplied by three.

Closure response: final-hour mean first-face viscosity falls from 1.32118 to 0.00417626 m^2/s ; heat diffusivity falls from 1.21593 to 7.01368e-05 m^2/s . Factor-3 viscosity is exactly zero at 98.473307% of sampled horizontal point-times, and heat diffusivity at 99.975586%. The timeline shows substantial initial mixing followed by mostly inactive coefficients, with intermittent reactivation. This is a change in the closure's operating regime, not simply a small uniform reduction in its strength.

Fraction interpretation: each hourly value averages 60 one-minute samples over all 32×32 horizontal points. Raw zero deficit, guard-valid zero deficit and actual zero coefficient are separately retained for momentum and heat; all six fractions appear in the detailed table. All three definitions agree in the final hour. Momentum and heat guards remain valid at 100% and 100%; coefficient-cap fractions are 0 and 0. Thus deficit satisfaction explains the switch-off, rather than invalid targets or active coefficient caps. Sampled zeros do not prove continuous switch-off between outputs. Factor 1 did not save these fractions; its values remain unavailable, not zero.

Moments and hourly sensitivity: factor-3 first-face w^3 is 0.00890504 m^3/s^3 and skewness 0.59496; peak w^2 is 0.0839403 and integrated resolved TKE 31.7259 m^3/s^2 . In 7-8 h, first-face w^2 is 0.0623155, w^3 0.00902423, skewness 0.580117, and integrated TKE 34.7786. The earlier-hour zero-viscosity/zero-heat-diffusivity fractions are 98.18034% / 99.96582%. The two neighboring windows show time dependence; they do not quantify ensemble uncertainty.

Factor 3 mostly switches the closure off

Transport and exchange: the SGS share of first-face u-momentum covariance-plus-SGS transport falls from 88.6586% with factor 1 to 1.14829% with factor 3. Factor-3 u^* is 0.272265 m/s and sensible heat flux -14.5912 W/m², versus 0.288603 m/s and -15.5382 W/m² for factor 1. Every plotted interior total is covariance plus actual constitutive SGS. It is NOT scheme-native WENO advective transport plus SGS: the numerical correction to covariance remains unmeasured. The factor experiment cannot identify or validate that missing flux.

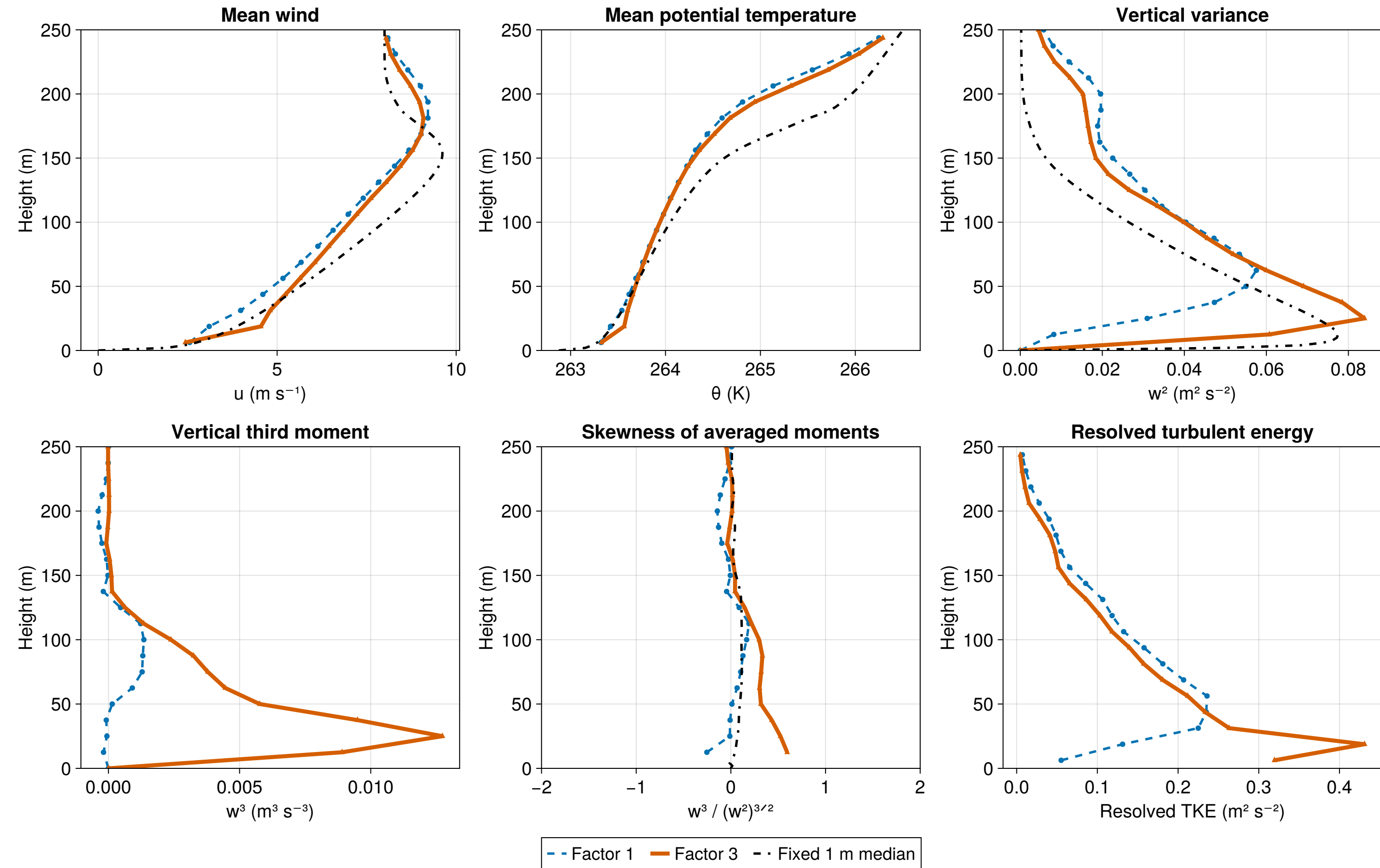
Reference comparison: at native heights $0 < z \leq 200$ m, factor-3 final-hour RMS errors against the fixed 1 m LES median are 0.720275 m/s for u , 0.376886 K for θ , and 0.0132648 m²/s² for w^2 . Changes relative to factor 1 are -23.19%, -9.761%, and -45.94%. The reference is an LES median, not observations. Greater resolved variance or a smaller mean-profile error alone does not establish faithful total transport.

Comparison integrity: factor 3 passed its source-specific strict admission (1 admitted, 0 rejected), with identical Breeze physics and initial native profiles, 19 exact profile times and 541 exact series times. Checksums, finite values, native vertical locations, one-face support, covariance-plus-SGS consistency and fraction bounds were verified. Factors 1/2/10 retain their original admissions. Factor 1 reproduces the corrected historical one-face300 output bit for bit; the no-closure case remains labeled as earlier-source context. The durable child record for non-login job 7348 verifies exit zero; no unavailable batch status is inferred.

Definitions: 8-9 h profiles average true half-hour means ending at 30600 and 32400 s; 7-8 h uses 27000 and 28800 s. Series averages exclude the hour start and include its end. Skewness is averaged w^3 divided by averaged w^2 to the three-halves power, masked for $w^2 < 1e-6$. Native model heights are retained; only the reference is interpolated for unweighted RMS errors. No scheme-native flux reconstruction, new closure design, or additional simulation is included.

Factor 3: mean profiles and resolved turbulence

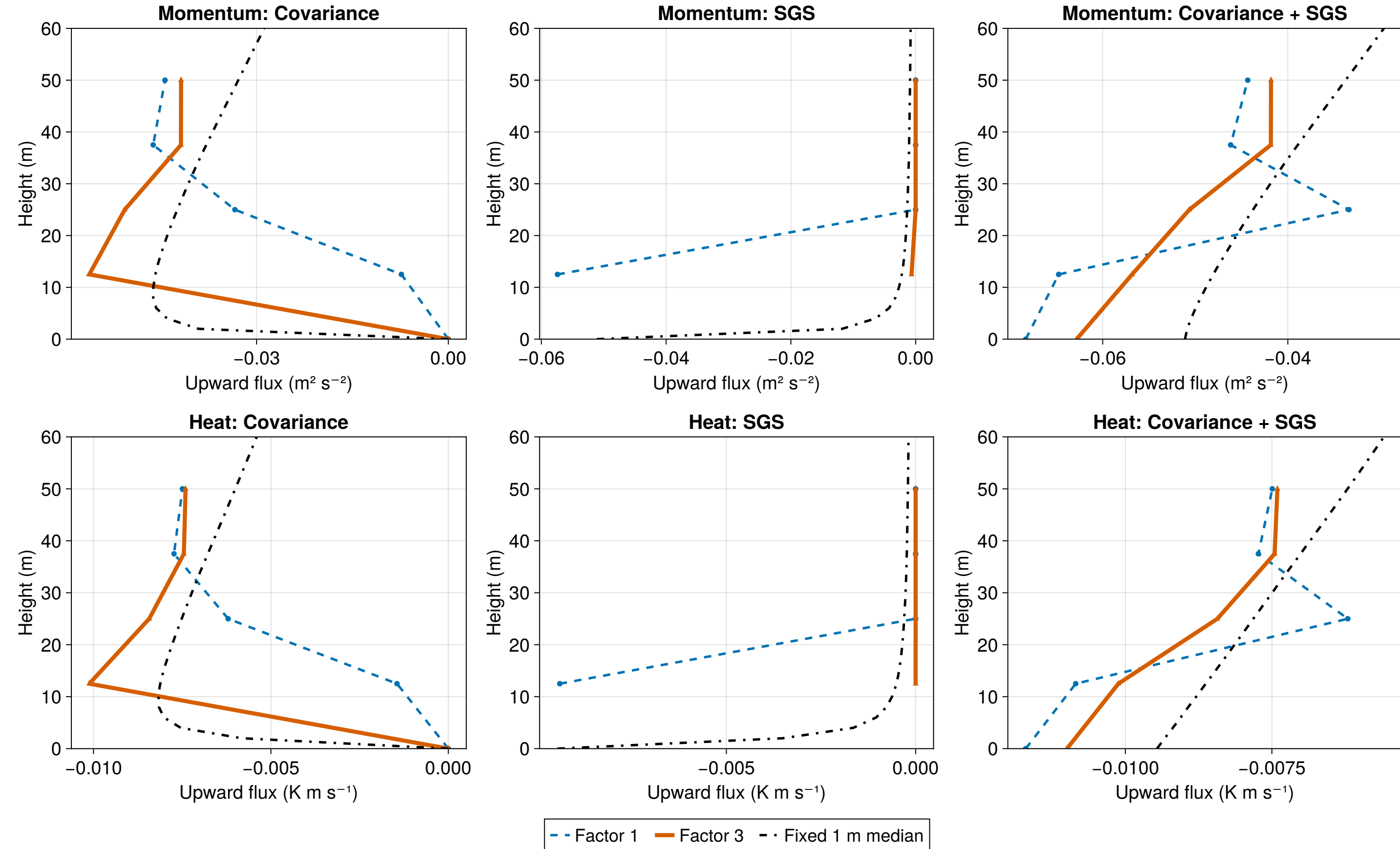
GABLS1 · 12.5 m · WENO9 · one face · 300 s filter · 8–9 h mean



Factor 1 is blue, dashed, circles; factor 3 is vermilion, solid, triangles.
Native heights and physical moments retained. Fixed 1 m reference medians shown where available.

Covariance and SGS transport near the surface

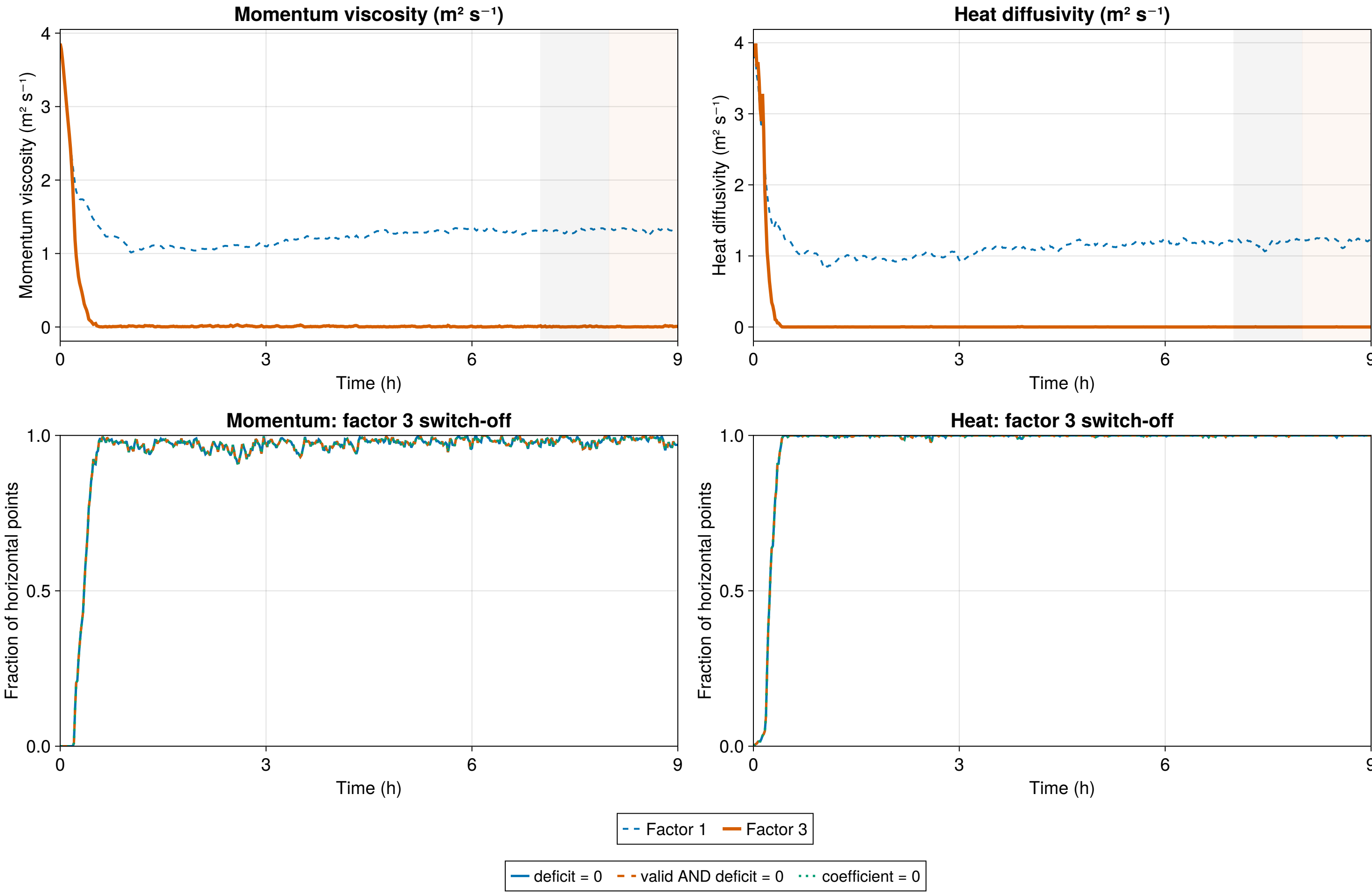
Same final-hour average · native flux faces · physical fluxes remain unscaled



Interior total = covariance + actual SGS. This is NOT scheme-native WENO transport: its numerical correction is missing.
Wall total is prescribed; artificial SGS wall zeros omitted. Fixed 1 m medians shown where available.

What the factor changes: coefficients and local switch-off

One supported face at 12.5 m · factors 1 and 3 · factor 1 switch-off fractions unavailable



All fraction denominators include every horizontal point. Exact overlap is meaningful; quantitative values are in the report.
Active/guard validity is separate from positive mixing. These six fractions were not saved for factor 1.